

Notes: Kehoe and Ruhl (JPE, 2013)

How Important Is the New Goods Margin in International Trade?

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1 Big picture

This paper studies whether growth in international trade is driven only by larger trade in already-traded goods, or also by the emergence of goods that were previously not traded, or traded only in very small amounts. Kehoe and Ruhl call this second channel the **new goods margin**, or the **extensive margin** of trade.

The key message is that the extensive margin matters especially during **trade liberalization** and **structural transformation**. When countries undergo major policy or economic changes, the composition of bilateral trade changes substantially: goods that were initially unimportant can become an economically large share of trade. In contrast, for stable country pairs without major trade-policy or structural change, the extensive margin changes very little.

2 One-sentence summary

This note summarizes Kehoe and Ruhl (2013), which develops a relationship-specific measure of the new-goods margin using five-digit bilateral commodity trade data, shows that least-traded goods grow strongly during liberalization and structural-transformation episodes such as NAFTA, China, Korea, and Chile, and argues that international trade models need mechanisms capable of generating changes in the set and composition of traded goods.

3 Research question

The paper asks three closely related questions:

1. How should researchers measure whether a good is newly traded or only marginally traded in bilateral trade data?
2. How important is the growth of least-traded goods for total trade growth?
3. Is the extensive margin mainly a long-run response to structural change, or does it also move substantially over normal business-cycle fluctuations?

The paper's answer is nuanced. The extensive margin is strongly associated with trade growth in broad panel data, but the economically large cases are concentrated in episodes where trade costs, production structure, or market access change persistently.

4 Incremental contribution

4.1 Contribution relative to prior extensive-margin measures

Earlier work often defines a good as nontraded using an absolute cutoff: zero recorded trade, or trade below a fixed dollar amount such as \$50,000. Kehoe and Ruhl argue that this approach is problematic because the same dollar cutoff has very different meanings for large and small trade relationships.

Their contribution is to define nontradedness **relatively**: a good belongs to the least-traded set if it is among the goods that collectively account for the bottom 10 percent of bilateral exports in the base period. This creates a cutoff that is specific to each exporter-importer pair.

4.2 Contribution relative to static cross-sectional evidence

Hummels and Klenow (2005) show that larger countries export a wider set of goods in the cross section. Kehoe and Ruhl provide a time-series analogue: when bilateral trade grows over time, especially during structural change, the least-traded goods grow disproportionately.

4.3 Contribution to quantitative trade modeling

The paper provides empirical discipline for models with fixed export costs, market-penetration costs, and endogenous product-market entry. The finding that the extensive margin is much stronger over long horizons than short horizons is consistent with models in which entry into export markets involves sunk costs and therefore responds more to persistent than transitory shocks.

5 Data and measurement

5.1 Data

The main data are annual bilateral commodity trade flows from the United Nations COMTRADE database. Goods are measured at the five-digit SITC revision 2 level, with 1,836 product codes. The broad panel contains 1,913 exporter-importer pairs over 1995-2005, representing more than 80 percent of world trade in the sample.

The paper also studies specific episodes:

- **Trade liberalization:** Canada-US FTA, NAFTA, and related North American trade flows.
- **Structural transformation:** China, Korea, and Chile.
- **Stable business-cycle episodes:** US trade with Germany, Japan, and the United Kingdom.

5.2 Least-traded goods measure

For each exporter-importer pair, the paper sorts goods by their average trade value over the first three years of the sample. Goods are then cumulated into ten groups, each accounting for 10 percent of base-period trade. The first group is the **least-traded goods** group. It includes all zero-trade goods and goods with very small trade values until the group reaches 10 percent of total exports.

This construction has two advantages. First, it avoids the instability of one-year zero trade observations caused by lumpy shipments. Second, it uses a relative threshold, making the measure comparable across small and large country pairs.

6 Empirical design and core specifications

6.1 Broad panel relationship

The paper first documents the relationship between total trade growth and growth of least-traded goods. For exporter m and importer n , the regression is summarized as

$$\Delta_{1995,t}^{1+k}(\hat{x}_{mn}) = \alpha + \beta \Delta_{1995,t}^{1+k}(x_{mn}) + \varepsilon_{mn}, \quad k = 1, \dots, 10,$$

where $\hat{x}_{mn}$ denotes trade in least-traded goods, and x_{mn} denotes total bilateral trade. The coefficient β is not interpreted causally; it is used to summarize how tightly extensive-margin growth and total-trade growth move together.

The headline broad-panel result is that over 1995-2005, a 10 percent increase in total trade is associated with a 35.9 percent increase in trade of least-traded goods using SITC data, and a 42.9 percent increase using Harmonized System data.

6.2 Episode analysis

The second empirical step studies specific country-pair episodes. This is important because the paper's theory-guided hypothesis is not simply that the extensive margin always grows, but that it grows most when structural fundamentals change persistently.

The comparison across episodes is central:

- In NAFTA and structural-transformation episodes, least-traded goods grow sharply.
- In stable US relationships with Germany, Japan, and the United Kingdom, least-traded goods remain close to their initial 10 percent share.

6.3 Trade-growth decomposition

The paper also decomposes total trade growth into intensive and extensive components. Let I_{t_0} be the set of goods traded in the initial period, I_{t_1} the set traded in the final period, and $I = I_{t_0} \cap I_{t_1}$ the set traded in both periods. The intensive-margin growth rate is

$$1 + g_{mn}^{IM} = \frac{\sum_{i \in I} x_{i,t_1}^{mn}}{\sum_{i \in I} x_{i,t_0}^{mn}}.$$

The extensive-margin component is defined as the residual contribution of the changing set of traded goods:

$$1 + g_{mn}^{EM} = \frac{\left(\sum_{i \in I_{t_1}} x_{i,t_1}^{mn}\right) / \left(\sum_{i \in I_{t_0}} x_{i,t_0}^{mn}\right)}{\left(\sum_{i \in I} x_{i,t_1}^{mn}\right) / \left(\sum_{i \in I} x_{i,t_0}^{mn}\right)}.$$

In logs, total trade growth is approximately decomposed as

$$g_{mn} \simeq g_{mn}^{EM} + g_{mn}^{IM}.$$

7 Main findings

7.1 Finding 1: The extensive margin is strongly related to long-run trade growth

Across 1,913 country pairs, least-traded goods grow much faster than total trade. The relationship strengthens as the horizon lengthens. One- and two-year relationships are much weaker, whereas the 10-year relationship is large and precise. This is consistent with a sunk-cost interpretation: firms and goods enter new markets in response to persistent changes, not small temporary shocks.

7.2 Finding 2: Trade liberalization activates the new goods margin

The NAFTA episode shows substantial growth in least-traded goods, especially for Canada-Mexico and Mexico-Canada trade. Canadian least-traded exports to Mexico rise from 10 percent of exports in 1989 to 30.3 percent in 1999, and Mexican least-traded exports to Canada rise to 25.3 percent.

7.3 Finding 3: Structural transformation produces even larger extensive-margin changes

China's transition, Korea's rapid industrialization, and Chile's transition to a more market-oriented economy are all associated with dramatic changes in the composition of trade. The most striking case is Korea-US trade, where least-traded Korean exports to the United States rise from 10 percent in 1975 to 65.8 percent in 1985.

7.4 Finding 4: Stable business-cycle episodes show little extensive-margin growth

US trade with Germany, Japan, and the United Kingdom shows little change in least-traded goods. This suggests that the extensive margin is not merely a mechanical time trend. It is most important when the economic environment changes persistently.

7.5 Finding 5: Fixed cutoff measures miss important changes

When the paper uses fixed cutoffs of zero or \$50,000 to define nontraded goods, the extensive margin often appears small or even negative. This is because large country pairs record positive trade in most goods, even if many goods are economically tiny. The paper's relative cutoff captures goods that are positive but small in the base period and become important later.

8 Identification and interpretation

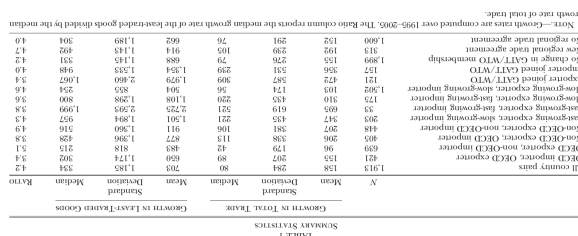
The paper is not a causal identification paper in the modern reduced-form sense. The authors explicitly avoid interpreting broad-panel coefficients causally. The design is instead descriptive and measurement-oriented, combining broad panel regularities with episode evidence.

The strength of the paper is that the timing and magnitude of extensive-margin growth differ systematically across episodes. The fact that the extensive margin rises around NAFTA and China's

The main limitation is that structural episodes are not clean natural experiments. China, Korea, and Chile changed many aspects of their economies at once, so the paper cannot isolate one policy shock. Instead, it shows that the new-goods margin is a major feature of such transformations.

9.1 Figure 1: Composition of exports - Canada to Mexico and Table 1: Summary statistics

Table 1: Summary statistics

TABLE 1
SUMMARY STATISTICS

Read: Table 1 summarizes 1,913 country pairs over 1995-2005. Least-traded goods grow much faster than total trade, and the ratios are especially large among fast-growing countries and GATT/WTO accession episodes.

Economic message: The new-goods margin is not a small anecdotal phenomenon. It appears systematically in a broad panel and is stronger in contexts associated with structural or policy change.

9.2 Figure 2: Least-traded goods and total trade growth and Table 2: Least-traded goods and total trade growth

Figure 2: Least-traded goods and total trade growth

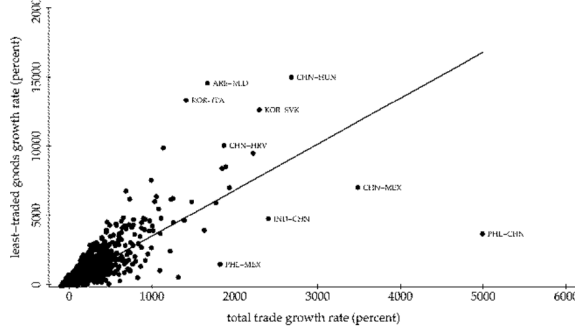


Table 2: Least-traded goods and total trade growth

HORIZON (Years)	SITC2		HARMONIZED SYSTEM 1992	
	Coefficient	R^2	Coefficient	R^2
1	.299** (.119)	.215	.498*** (.053)	.413
2	.493*** (.044)	.403	.704*** (.041)	.595
3	2.016*** (.192)	.430	1.885*** (.151)	.525
4	2.246*** (.129)	.468	2.439*** (.123)	.497
5	2.836*** (.232)	.503	2.735*** (.220)	.566
6	2.425*** (.381)	.517	2.829*** (.245)	.564
7	3.108*** (.234)	.663	3.205*** (.323)	.686
8	3.752*** (.385)	.682	4.241*** (.426)	.791
9	3.599*** (.330)	.725	3.647*** (.387)	.757
10	3.591*** (.402)	.717	4.288*** (.243)	.837
Observations	1,913		1,277	

NOTE.—Ordinary least squares estimates of eq. (1). Each row reports the coefficient and R^2 for a specification of (1) in which the growth rates are taken over the number of years listed in the column Horizon. An observation is an exporter-importer pair. Robust standard

Read: Figure 2 plots least-traded goods growth against total trade growth for the broad country-pair sample. The relationship is strongly positive, even though there are outliers.

Economic message: Country pairs with larger increases in total trade also tend to experience much larger increases in initially marginal goods.

Read: Table 2 shows that the coefficient linking total trade growth to least-traded-goods growth rises with the horizon. In SITC data it grows from 0.299 at one year to 3.591 at ten years.

Economic message: The extensive margin is primarily a long-run margin. This supports models where entering export markets takes time and responds more to persistent changes than temporary fluctuations.

9.3 Figure 3: Composition of exports - Mexico to Canada and Figure 4: Least-traded goods - Mexico and Canada

Figure 3: Composition of exports - Mexico to Canada

findings regarding the NAFTA are broadly consistent with those of Romalis (2007), who finds that the NAFTA had the largest impact on trade between Mexico and Canada. Additionally, Romalis finds that the goods

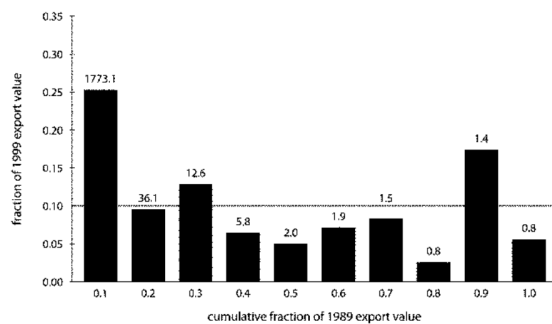


FIG. 3.—Composition of exports: Mexico to Canada

Figure 4: Least-traded goods - Mexico and Canada



FIG. 4.—Least-traded goods: Mexico and Canada

Read: Mexico's least-traded exports to Canada rise from 10 percent in 1989 to 25.3 percent in 1999. The spike is again concentrated among initially small or nontraded categories.

Economic message: The Canada-Mexico liberalization changed trade composition in both directions, not only trade volume.

Read: The time-series paths for Canada-to-Mexico and Mexico-to-Canada least-traded goods rise sharply around and after the implementation of NAFTA.

Economic message: Timing matters: the increase occurs around the trade-policy change rather than gradually drifting upward in unrelated periods.

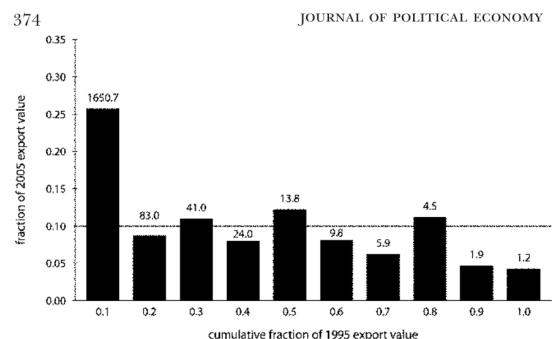
9.4 Table 3: Share of least-traded goods - trade liberalization and Figure 5: Composition of exports - China to United States

Table 3: Share of least-traded goods - trade liberalization

outlined here, Mukerji (2009) finds that, during unilateral trade liberalization in India, the least-traded goods grew from 10 percent to 33.8 percent of total imports over the period 1988–99 and from 10 percent to 26.5 percent of total exports. In another study, also using our methodology, Sandrey and van Severen (2004) find that, during liberalization of

Period	Trade Flow	Share of Total Exports
1989–99	Canada to Mexico	.303
1989–99	Mexico to Canada	.253
1988–98	Canada to United States	.166
1988–98	United States to Canada	.121
1989–99	Mexico to United States	.180
1980–88	Mexico to United States	.357
1989–99	United States to Mexico	.137
1980–88	United States to Mexico	.223

Figure 5: Composition of exports - China to United States



Read: Table 3 reports end-period least-traded-goods shares for North American liberalization. Canada-Mexico and Mexico-Canada show the largest NAFTA-period shifts.

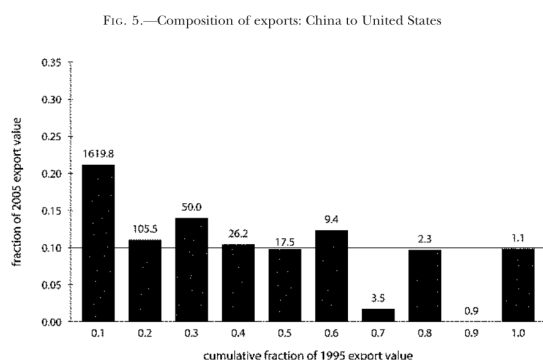
Economic message: Trade liberalization produces the strongest new-goods response when the initial bilateral relationship was relatively underdeveloped.

Read: China's least-traded exports to the United States in 1995 account for 25.7 percent of exports by 2005.

Economic message: China's trade growth is not merely "more of the same." It reflects a broadening of the product set exported to the United States.

9.5 Figure 6: Composition of exports - United States to China and Figure 7: Least-traded goods - China and United States

Figure 6: Composition of exports - United States to China



Read: US least-traded exports to China also grow substantially, reaching 21.2 percent of exports by 2005.

Economic message: Structural change in China affects both Chinese exports and imports. As China opens and industrializes, US exporters also begin selling a wider set of goods.

Figure 7: Least-traded goods - China and United States



Read: Figure 7 shows different timing: US exports to China rise strongly in the late 1990s, while Chinese exports to the US rise especially after 2000.

Economic message: The timing aligns with different stages of China's reform and WTO accession, suggesting that the extensive margin responds to major institutional and market-access shifts.

9.6 Table 4: Share of least-traded goods - structural transformation and Figure 8: Least-traded goods growth over longer time horizon

Table 4: Share of least-traded goods - structural transformation

TABLE 4 SHARE OF LEAST-TRADED GOODS: STRUCTURAL TRANSFORMATION		
Period	Trade Flow	Share of Total Exports
1975-85	Chile to United States	.346
1975-85	United States to Chile	.707
1995-2005	China to United States	.257
1995-2005	United States to China	.212
1975-85	Korea to United States	.658
1975-85	United States to Korea	.560

traded set of goods in 1975. These new exports include Korea's second-largest export in 1985, electronic microcircuits (code 77640), as well as electrothermic domestic appliances (code 77586). This evidence suggests

Figure 8: Least-traded goods growth over longer time horizon



FIG. 8.—Least-traded goods growth over longer time horizon

Read: Table 4 shows very large extensive-margin shares in structural transformation episodes: Korea-to-US reaches 65.8 percent, US-to-Korea 56.0 percent, and US-to-Chile 70.7 percent.

Economic message: Structural transformation can remake the product composition of trade, not just scale up preexisting exports.

Read: Figure 8 extends the horizon and shows that least-traded-goods shares can keep rising beyond the initial episode window.

Economic message: Entry and product diversification are gradual processes. The extensive margin may continue to evolve after formal policy changes.

9.7 Table 5: Share of least-traded goods - business-cycle fluctuations and Figure 9: Composition of exports - United Kingdom to United States

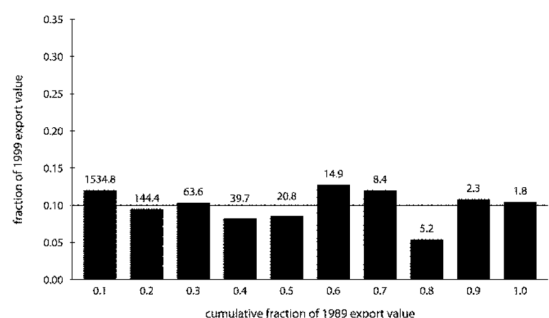
Table 5: Share of least-traded goods - business-cycle fluctuations

these three have had no major trade relation with the United States, nor were they part of a structural transformation episode over the sample period. We have already examined the extensive margin for the other four major trading partners of the United States: Canada, Mexico, China, and Korea.

As can be seen in table 5, none of these relationships is characterized by the large growth in the extensive margin that we saw in tables 3 and 4.

TABLE 5 SHARE OF LEAST-TRADED GOODS: BUSINESS CYCLE FLUCTUATIONS		
Period	Trade Flow	Share of Total Exports

Figure 9: Composition of exports - United Kingdom to United States



Read: Table 5 shows that stable US trade relationships with Germany, Japan, and the UK have end-period least-traded-goods shares close to 10-15 percent, much smaller than the liberalization and transformation episodes.

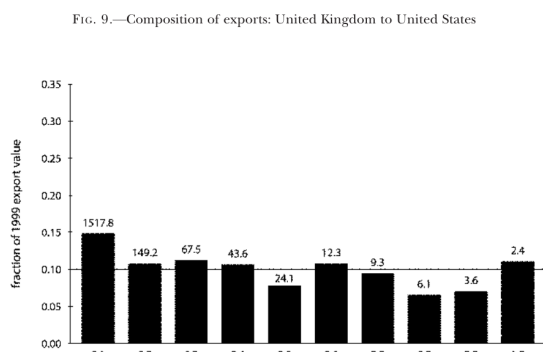
Economic message: The extensive margin does not automatically explode over a decade. It is tied to structural change, not merely the passage of time.

Read: UK-to-US exports show bars clustered around the initial 10 percent benchmark, with no dramatic spike in the least-traded group.

Economic message: In a mature and stable trade relationship, trade growth is mostly intensive rather than extensive.

9.8 Figure 10: Composition of exports - United States to United Kingdom and Figure 11: Least-traded goods - United States and United Kingdom

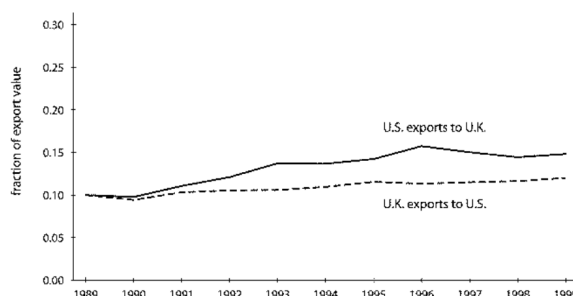
Figure 10: Composition of exports - United States to United Kingdom



Read: US-to-UK exports similarly show little compositional reshuffling relative to the trade-liberalization and structural-transformation cases.

Economic message: Stable country pairs provide an important benchmark: without persistent shocks, the new-goods margin remains modest.

Figure 11: Least-traded goods - United States and United Kingdom



Read: Figure 11 shows smooth and small changes in the least-traded-goods share for US-UK trade.

Economic message: Normal business-cycle fluctuations are not enough to generate large product-entry responses. This supports sunk-cost and persistence-based interpretations.

9.9 Table 6: Decomposing total trade growth and Figure 12: Canada to Mexico - alternative sorting procedure

Table 6: Decomposing total trade growth

each growth rate is the contribution of that margin to total trade growth. For example, total exports from Canada to Mexico grew by 71.6 percent over 1989–99. The extensive margin accounted for 16.7 percent of the growth in total trade, and the intensive margin accounted for the other 83.3 percent.

The patterns we have found in the previous sections are borne out in the decompositions as well: the extensive margin is a significant con-

	LOG DIFFERENCE		
	Total Trade (1)	Extensive Margin (2)	Intensive Margin (3)
Canada to Mexico, 1989–99	.716	.120	.597
		[16.7]	[83.3]
Mexico to Canada, 1989–99	1.565	.097	1.468
		[6.2]	[93.8]
Canada to United States, 1988–98	.788	.092	.696
		[11.7]	[88.3]
United States to Canada, 1988–98	.706	.061	.645
		[8.7]	[91.3]
Mexico to United States, 1989–99	1.486	.136	1.350
		[9.2]	[90.8]
United States to Mexico, 1989–99	1.310	.088	1.222
		[6.7]	[93.3]
Chile to United States, 1975–85	1.962	.033	1.929
		[1.7]	[98.3]
United States to Chile, 1975–85	.987	.834	.152
		[84.6]	[15.4]
China to United States, 1995–2005	1.683	.258	1.424
		[15.4]	[84.6]
United States to China, 1995–2005	1.308	−.113	1.421
		[−8.6]	[108.6]
Korea to United States, 1975–85	3.157	1.015	2.143
		[32.1]	[67.9]
United States to Korea, 1975–85	1.892	.580	1.312
		[30.7]	[69.3]
Germany to United States, 1989–99	.842	.086	.756
		[10.2]	[89.8]
United States to Germany, 1989–99	.741	−.017	.759
		[−2.3]	[102.3]
Japan to United States, 1989–99	.342	.002	.340
		[.6]	[99.4]
United States to Japan, 1989–99	.375	.003	.373
		[.8]	[99.2]

Read: Table 6 decomposes total trade growth into extensive and intensive components. NAFTA country pairs average a 9.9 percent extensive-margin contribution, structural episodes average 26.0 percent, and stable pairs average only 2.4 percent.

Economic message: The extensive margin is economically meaningful in the right environments, but it does not dominate everywhere.

Figure 12: Canada to Mexico - alternative sorting procedure

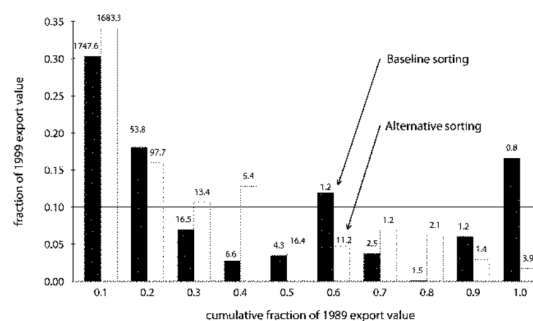


FIG. 12.—Canada to Mexico: alternative sorting procedure

Read: Figure 12 compares the baseline sorting method with sorting by trade relative to world trade for Canada-to-Mexico exports. The least-traded share remains large under both methods.

Economic message: The main NAFTA results are not driven by arbitrary product-code width or the baseline sorting convention.

9.10 Figure 13: Mexico to Canada - alternative sorting procedure and Table 7: Decomposition under different cutoff values

Figure 13: Mexico to Canada - alternative sorting procedure

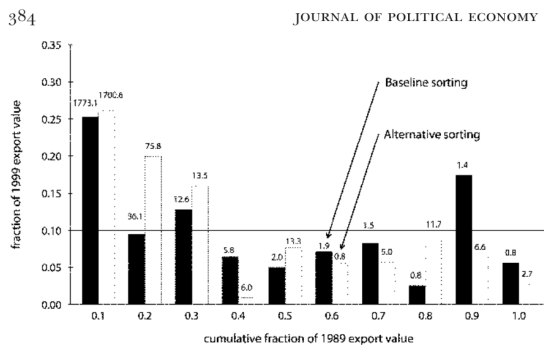


FIG. 13.—Mexico to Canada: alternative sorting procedure

Table 7: Decomposition under different cutoff values

	LOG DIFFERENCE			CUTOFF VALUE (4)
	Total Trade (1)	Extensive Margin (2)	Intensive Margin (3)	
Canada to Mexico, 1989–99	.642	–.049 [–7.7]	.691 [107.7]	0
	.648	–.038 [–5.9]	.686 [105.9]	50
	.716	.120 [16.7]	.597 [83.3]	481
Mexico to Canada, 1989–99	1.502	–.077 [–5.2]	1.579 [105.2]	0
	1.504	–.019 [–1.3]	1.523 [101.3]	50
	1.565	.097 [6.2]	1.468 [93.8]	2,529
Canada to United States, 1988–98	.742	.001 [.2]	.740 [99.8]	0
	.742	.001 [.1]	.741 [99.9]	50
	.788	.092 [11.7]	.696 [88.3]	40,932
United States to Canada, 1988–98	.656	.007 [1.1]	.649 [98.9]	0
	.656	.008 [1.2]	.649 [98.8]	50
	.706	.061 [8.7]	.645 [91.3]	26,322
Mexico to United States, 1989–99	1.412	.062 [4.4]	1.350 [95.6]	0
	1.412	.065 [4.6]	1.348 [95.4]	50
	1.486	.136 [9.2]	1.350 [90.8]	19,634
United States to Mexico, 1989–99	1.234	.028 [2.3]	1.207 [97.7]	0
	1.235	.027 [2.2]	1.207 [97.8]	50
	1.310	.088 [6.7]	1.222 [93.3]	8,838

NOTE.—Numbers in brackets are the contribution to the total. Cutoff values are in thousands of US dollars.

traded goods, the $\bar{x}^m$ from Section IV. In contrast to the other cutoffs, this value varies across country pairs. Column 1 of the table reports the growth in total trade between the country pair. Note that the growth rate of total trade differs slightly across the different cutoffs because a good that

Read: Figure 13 shows that the alternative sorting procedure produces similar, if slightly different, evidence for Mexico-to-Canada exports.

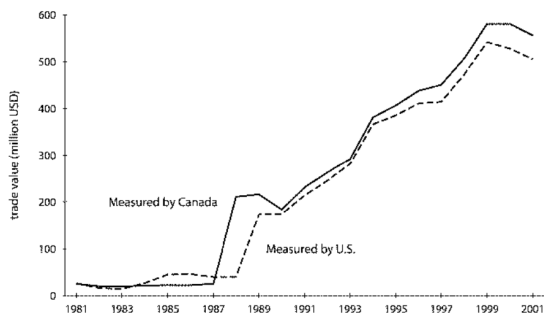
Economic message: The measured extensive-margin response is robust to normalizing trade by world trade in each good.

Read: Table 7 compares zero, \$50,000, and relationship-specific cutoffs. Fixed cutoffs often imply little or negative extensive-margin growth, while the authors' relative cutoff captures economically meaningful changes.

Economic message: Measurement is central. For large trade flows, nearly all products can have positive recorded trade, so zero-based measures miss growth in initially small goods.

9.11 Figure A1: SITC 74930 and the switch to the Harmonized System and Table A1: Code turnover

Figure A1: SITC 74930 and the switch to the Harmonized System



Read: Figure A1 shows a sharp jump in measured Canadian transmission parts around the adoption of the Harmonized System, with the timing differing by reporting country.

Economic message: Some apparent new-goods growth can be an artifact of classification changes. Careful sample-period selection is necessary.

Table A1: Code turnover

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TABLE A1
CODE TURNOVER

	A. US IMPORTS FROM CANADA				B. US EXPORTS TO CANADA			
	US-Collected		Canadian-Collected		US-Collected		Canadian-Collected	
	Births	Deaths	Births	Deaths	Births	Deaths	Births	Deaths
1982	0	4	4	2	41	43	5	2
1983	3	4	4	2	43	38	4	4
1984	5	13	2	0	49	20	3	0
1985	87	36	1	2	70	74	1	2
1986	4	23	0	1	36	26	0	1
1987	8	9	2	1	34	40	2	1
1988	13	6	996	11	43	32	1,167	16
1989	364	105	117	78	325	152	132	96
1990	5	15	76	32	38	49	91	36
1991	122	15	26	29	152	66	28	37
1992	4	17	28	26	52	39	36	32
1993	13	6	29	27	38	45	34	33
1994	7	8	28	31	48	48	35	41
1995	9	9	29	28	48	38	40	33
1996	11	6	29	22	42	35	36	27
1997	5	7	29	22	36	34	34	25
1998	7	10	24	21	40	35	28	30
1999	6	5	22	34	38	46	29	40
2000	6	11	29	27	37	57	33	34
2001	4	6	17	20	39	36	23	22
Average	17.1	13.6	2.2	1.3	45.1	39	2.5	1.7

NOTE.—Average is the average for that column over the years preceding the transition

Read: Table A1 documents large code births and deaths around 1988-1989, exactly when Canada and the United States adopted the Harmonized System.

Economic message: The appendix validates the authors' decision to avoid crossing classification transition years when measuring the extensive margin.

10 Short summary

- The paper measures the extensive margin using goods that account for the bottom 10 percent of base-period bilateral trade.
- Least-traded goods grow disproportionately when total trade grows over long horizons.
- The extensive margin is especially important during trade liberalizations and structural transformations.
- Stable business-cycle episodes show little change in the extensive margin.
- Fixed cutoffs such as zero trade or \$50,000 miss much of the relevant product-composition change.

11 Conclusion

11.1 Core conclusion

The new-goods margin is an important component of international trade growth, but mainly when the underlying economic environment changes persistently. Trade liberalization and structural

transformation induce countries to begin exporting or importing goods that were previously absent or quantitatively tiny.

11.2 Economic intuition

Entering a new export relationship or scaling a tiny good into a meaningful trade flow likely requires fixed or sunk investments: search, distribution, regulatory compliance, product adaptation, and customer acquisition. Such investments are more likely when trade reforms or structural transformations permanently improve expected profitability.

11.3 Why the paper matters

The paper challenges models with fixed production and export structures. If a model cannot generate changes in the set of traded goods, it will miss an important part of how trade responds to liberalization and development. This is why the paper speaks directly to heterogeneous-firm models, market-penetration-cost models, and quantitative trade models of policy reform.

11.4 Main limitation

The paper is primarily descriptive and measurement-focused. It does not provide a single causal estimate of one reform. However, its careful comparison across liberalization, structural transformation, and stable episodes makes a persuasive case that extensive-margin growth is a key feature of large, persistent trade changes.

11.5 Takeaway

The extensive margin is not just about whether a product has exactly zero recorded trade. It is about whether initially marginal goods become important. Measuring that transformation requires a relationship-specific notion of tradedness, and Kehoe and Ruhl provide one of the cleanest tools for doing so.